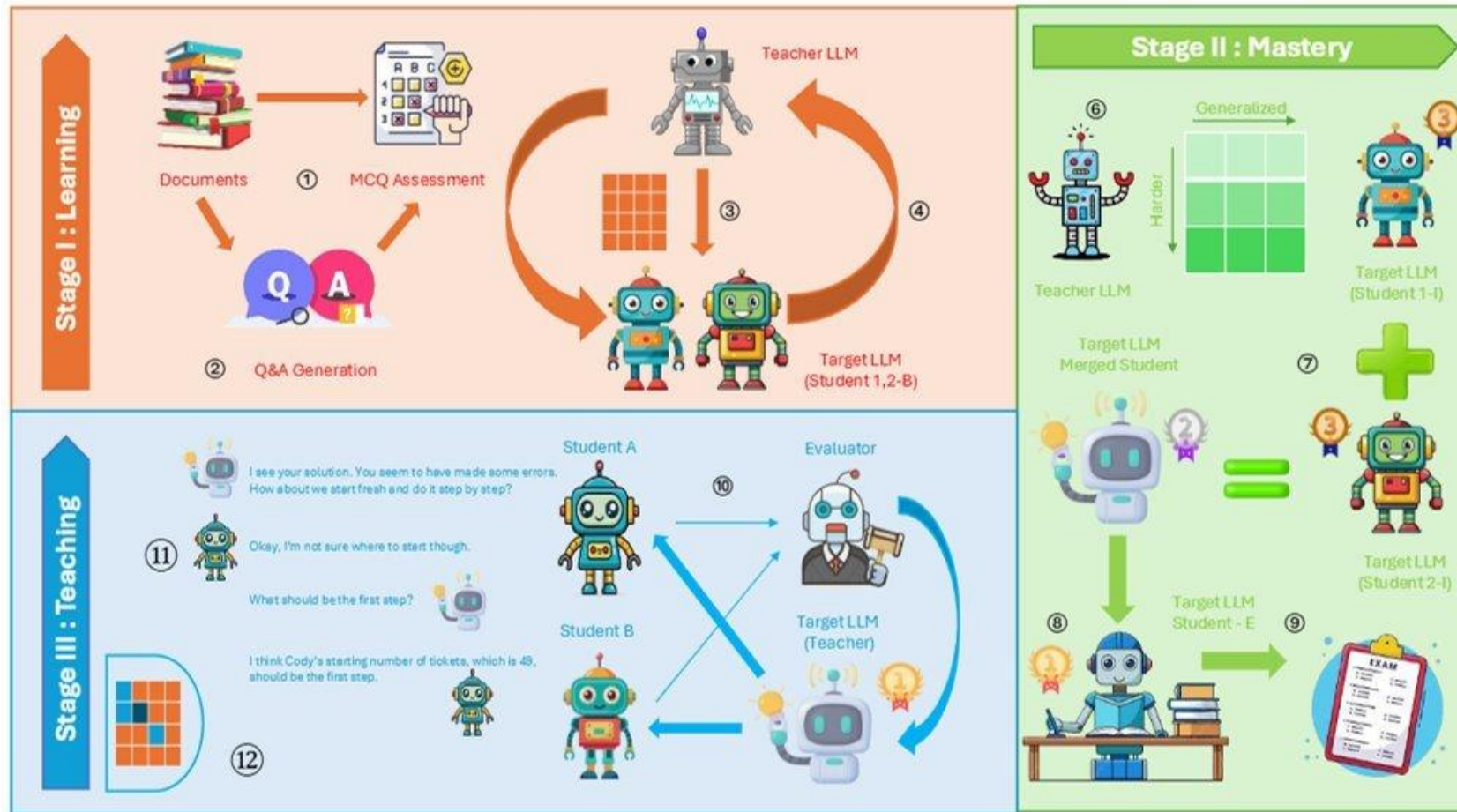
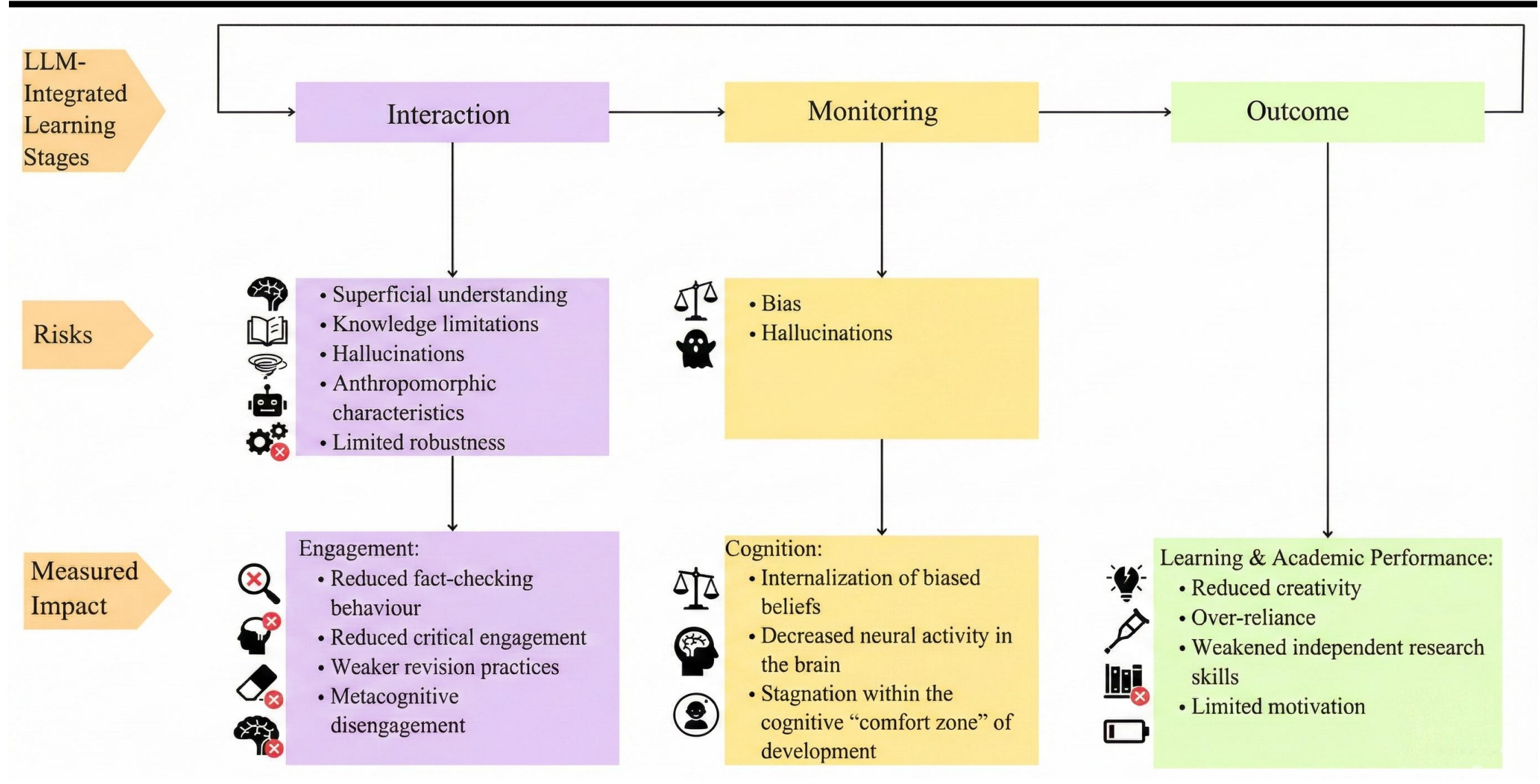


MENTORX

An AI teacher that can learn like a student, reflect like a researcher, and teach like a professor.



PROBLEM STATEMENT



KNOWLEDGE GAP ASSESSMENT + TEACHING MATERIAL CURATION AN OVERVIEW



An automated teaching-material curator that turns a single source document into a strategic topic plan, a knowledge graph, MCQs with distractors, and an adaptive bulk dataset with key focus on weak learner topics.



Solves scalable curriculum + assessment creation by simulating a student, detecting hard concepts, and reallocating question generation toward those weak areas.



A Strategic-Plan crew ingests the document, applies RACAR-style scoring (relevance, alignment, coverage, atomicity, reasoning), and normalizes everything into JSON topic plans for downstream agents on how to ask question on a single topic of a chapter.



A Knowledge-Graph crew generates and enriches DOT graphs, infers relationship metadata, and adds visual labels for Graphviz-based rendering and inspection.



MCQ and distractor generators crew generates questions which enforce Bloom's Taxonomy and are a mix of easy, medium and hard based on the knowledge graph.



An adaptive bulk-dataset module allocates per-topic question counts via weak/strong splits, micro-batching, dynamic timeouts, and a top-up pass until target volume is reached.

KNOWLEDGE GAP ASSESSMENT + TEACHING MATERIAL CURATION

KNOWLEDGE GRAPH & ITERATIVE MULTI-AGENT REFINEMENT



Each topic in our system is designed to understand:

- What level of thinking it requires
- What skills it tests
- How the concept is represented (text, diagram, math, etc.)
- Common mistakes learners usually make
- Key formulas that connect ideas
- This ensures every learning unit is **pedagogy-aware, not just content-aware**.



Topics don't stand alone — they are connected by:

- How strongly they are related
- What mental operation is needed (apply, analyze, evaluate)
- How difficult the jump is
- Typical misconceptions during the transition
- Which formula links the two ideas
- This creates a **true learning pathway**, not just a topic list.



Every major module follows a 3-step refinement process:

- Generate** – Create the content
- Review** – Critically evaluate quality & alignment
- Refine** – Improve accuracy, difficulty, and clarity
- This applies to:
 - Learning plans
 - Questions & answers
 - Distractors (wrong options based on real mistakes)
 - Nothing moves forward without passing quality checks



The system:

- Tracks where learners perform well or struggle
- Identifies strong vs weak topics
- Allocates more questions to weaker areas
- Automatically balances the final dataset size
- Refines again if needed
- This creates a **personalized, performance-driven question bank**.

STRUCTURE-AWARE LEARNING-FROM-TEACHER (LFT)



Goal:

Develop a CBSE-math-specialized Llama-3.1-8B model that reproduces teacher-style reasoning using chain-of-thought and knowledge-graph structure.



Approach:

PromptKD: Prompt-aligned teacher–student distillation

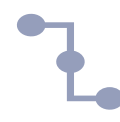
StructTuning: Structure-aware supervision (concept nodes, edges, operations, representation paths)

Curriculum: Chapter-wise training over **35 chapters** (~17,500 structured examples)



Output:

Merged LFT-finetuned model hosted on Hugging Face
(Llama-3.1-8B base + LFT adaptations combined into a single checkpoint)



Key Ideas:

Not just fine-tuning:
a **concept-transfer pipeline**

Model learns conceptual structure rather than memorizing solutions

Provides the structured reasoning backbone for the ICR (Interactive Concept Refinement) module

LFT RESULTS & WHAT THE MODEL LEARNS



Performance Improvements (Structured Evaluation: 2,617 Questions)

Base Model Accuracy: 23.88%
LFT Fine-Tuned Accuracy: 32.21%
Overall Gain: +8.33 percentage points
Correct answers: 625 → 843



Text Similarity Metrics (Explanation Alignment; Finetuned Model Metric / Base Model Metric)

BLEU Score :- 58.56 / 38.24
ROUGE Score :- 0.4185 / 0.2938
METEOR Score :- 0.2327 / 0.1633



What LFT Actually Teaches the Model

Better alignment with concept nodes,
edges, operations
More stable and consistent teacher-style
chain-of-thought
Improved mapping from question →
conceptual structure
Reduced conceptual drift &
hallucinations
Stronger performance on concept-
heavy chapters (e.g., 3D Geometry,
Coordinate Geometry)



Outcome

The merged LFT model forms a
conceptually stable, structure-aware
foundation, enabling the Interactive
Concept Refinement (ICR) module to
work effectively.

INTERACTIVE CONCEPT REFINEMENT (ICR): MOTIVATION & CORE IDEA



Why ICR?

LFT teaches structured reasoning, but it **does not teach the model how to learn interactively**.



Real students improve when they:

- attempt a solution
- become unsure about a reasoning step
- ask doubt/clarification
- receive conceptual feedback
- revise reasoning

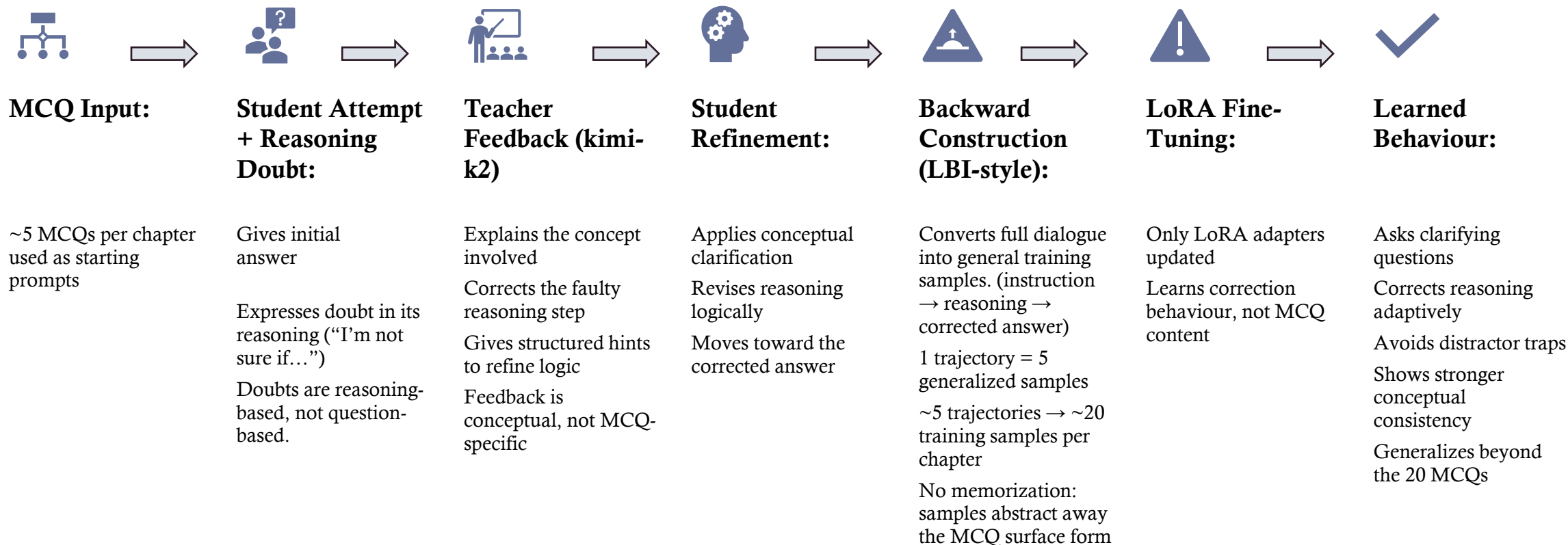
ICR adds exactly this **interactive improvement behaviour**.



Core Idea

Simulate doubt-clearance sessions:
Student model attempts the MCQ
Student expresses **doubt about its reasoning**, not the MCQ content
Teacher model (kimi-k2) provides **conceptual clarification**, identifies wrong reasoning steps
Student refines reasoning based on guidance
Full dialogue is converted into **general training samples** via Backward Construction
Fine-tuning with **LoRA** teaches the model how to self-correct

ICR PIPELINE



INTERACTIVE CONCEPT REFINEMENT RESULTS



Performance Improvements (Structured Evaluation: 2,617 Questions)

LFT-finetuned Model Accuracy: 32.21%
After ICR Fine-Tuning: 35.00%
ICR Gain: +2.79 percentage points



Text Similarity Metrics (Explanation Alignment; LFT+ICR Finetuned Model Metric / LFT Finetuned Model Metric)

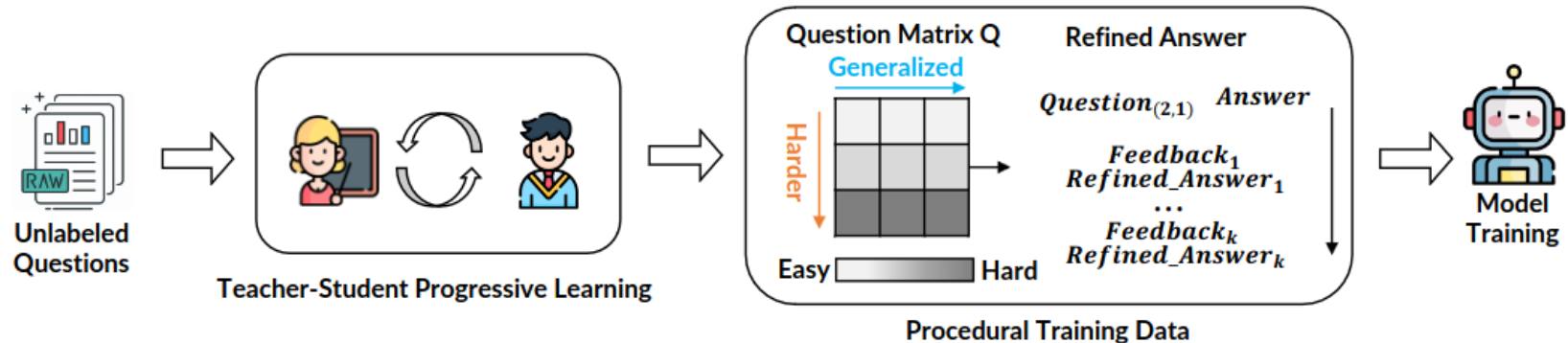
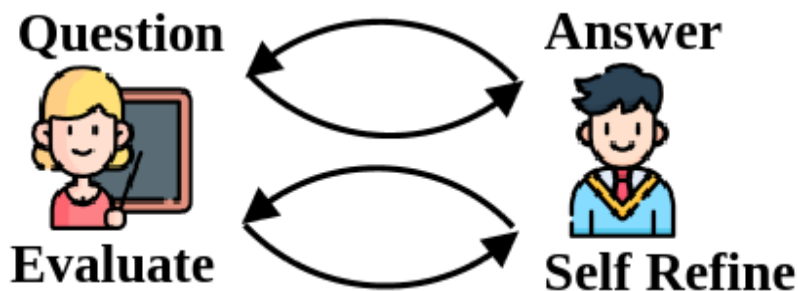
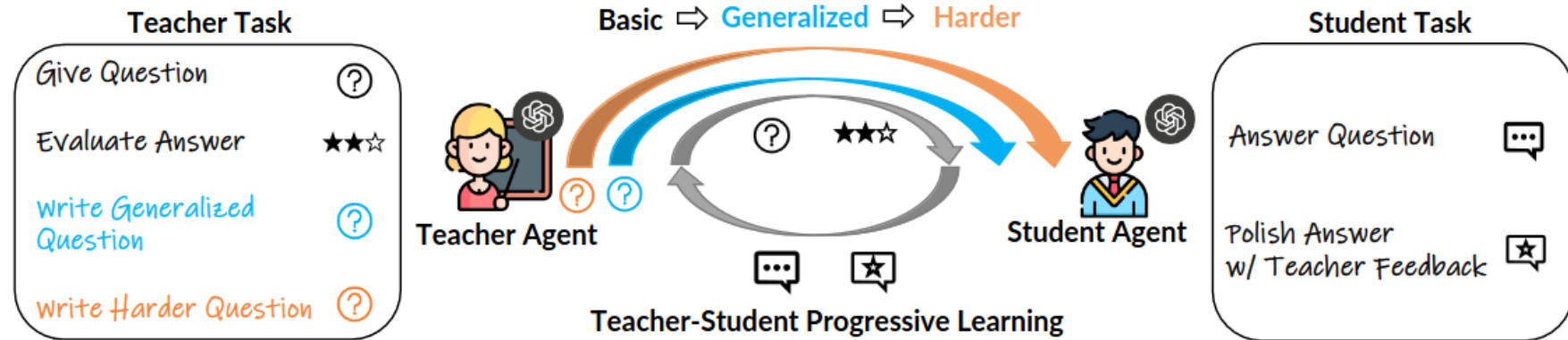
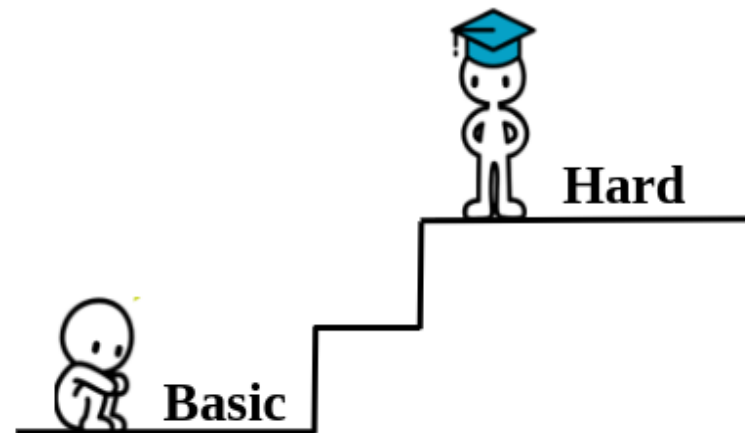
BLEU score: 59.31/58.56
ROUGE-L score: 0.4414/0.4181
METEOR score: 0.2478/0.2327



Outcome

ICR improves conceptual reliability and interactive learning behaviour, producing a model that not only solves but **learns through feedback**, just like a real student.

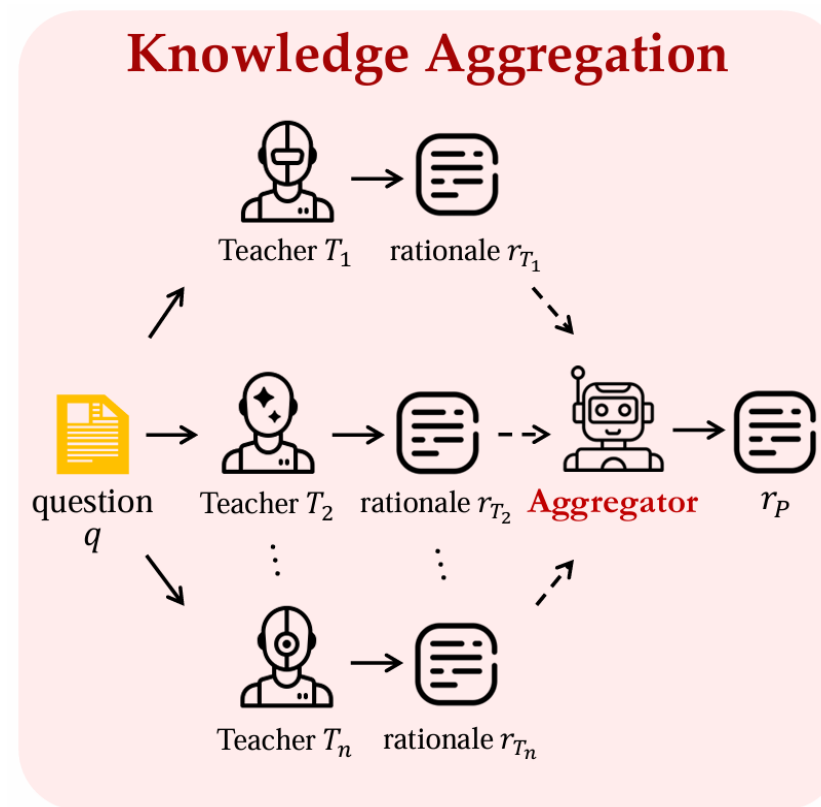
STAGE V : FEEDBACK-DRIVEN PRACTICE



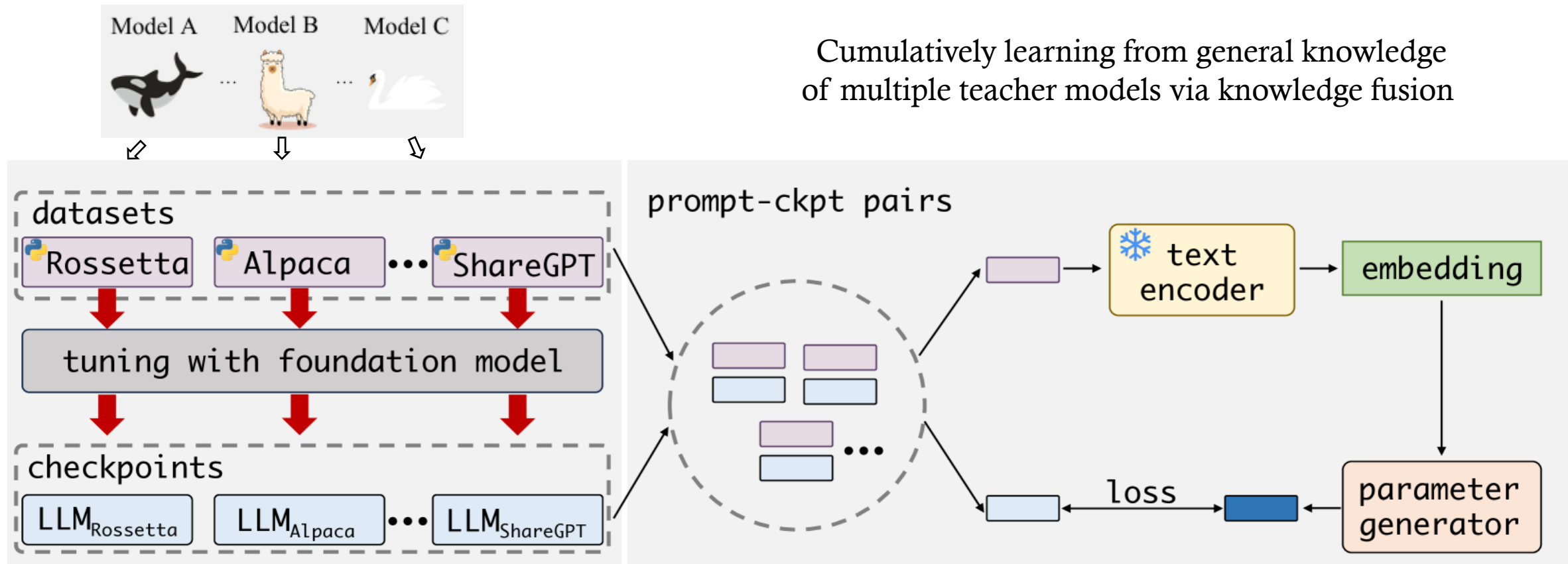
Cumulative Learning Experience

Why Cumulative Learning is Crucial ?

- Students learning from a single source face the issue of **creativity** and **diversity** in problem solving
- Such students also face the potential risk of **overfitting** to the **biases** or **inconsistencies** of one teaching pattern
- To mitigate these, it's crucial for students to learn from multiple teaching sources so that they are exposed to a wide variety of reasoning approaches which will subsequently enhance their intrinsic reasoning aptitudes as well

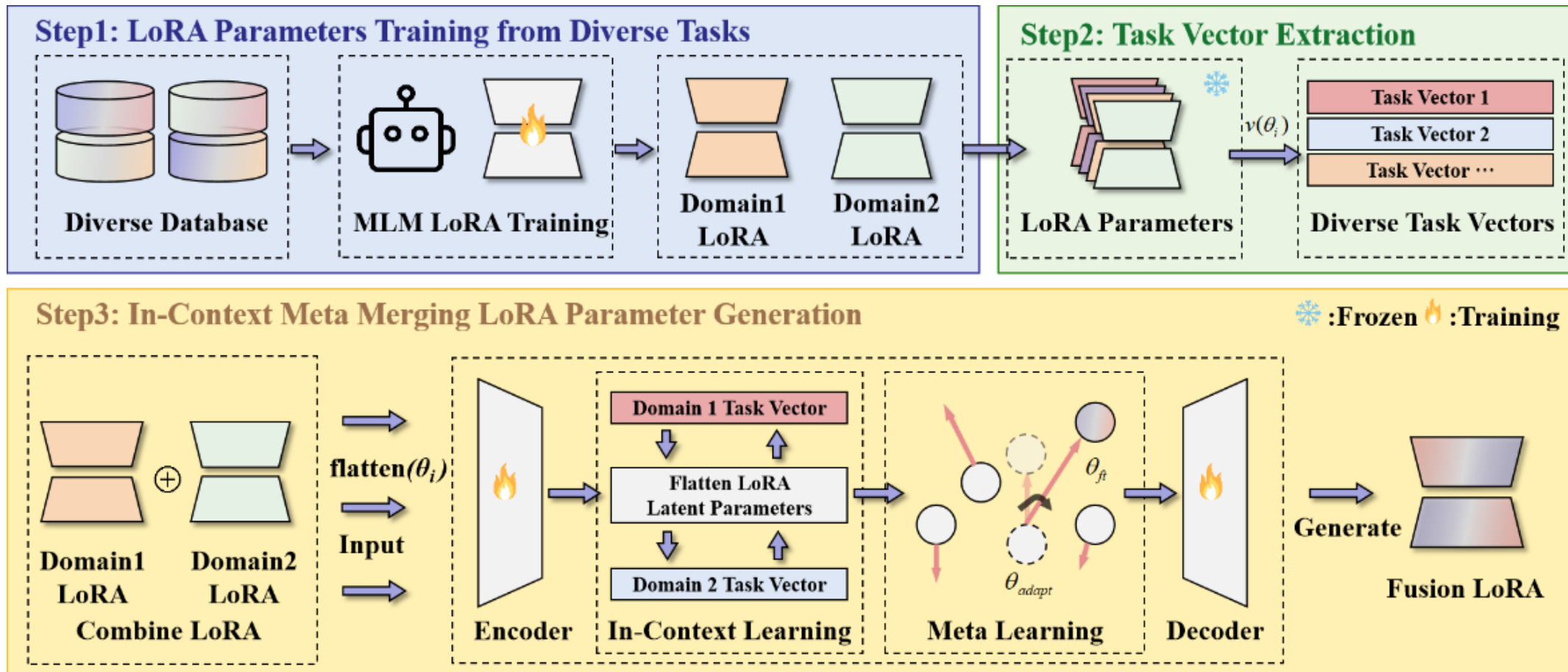


Cumulative Learning Experience



Stage 1 : Training a subject-specific meta-student. This process involves first collecting datasets from multiple teacher models (A, B, C) and then fine-tuning our student on all these individually. This is then facilitated by a fusion to aggregate all the knowledge into one student

Cumulative Learning Experience

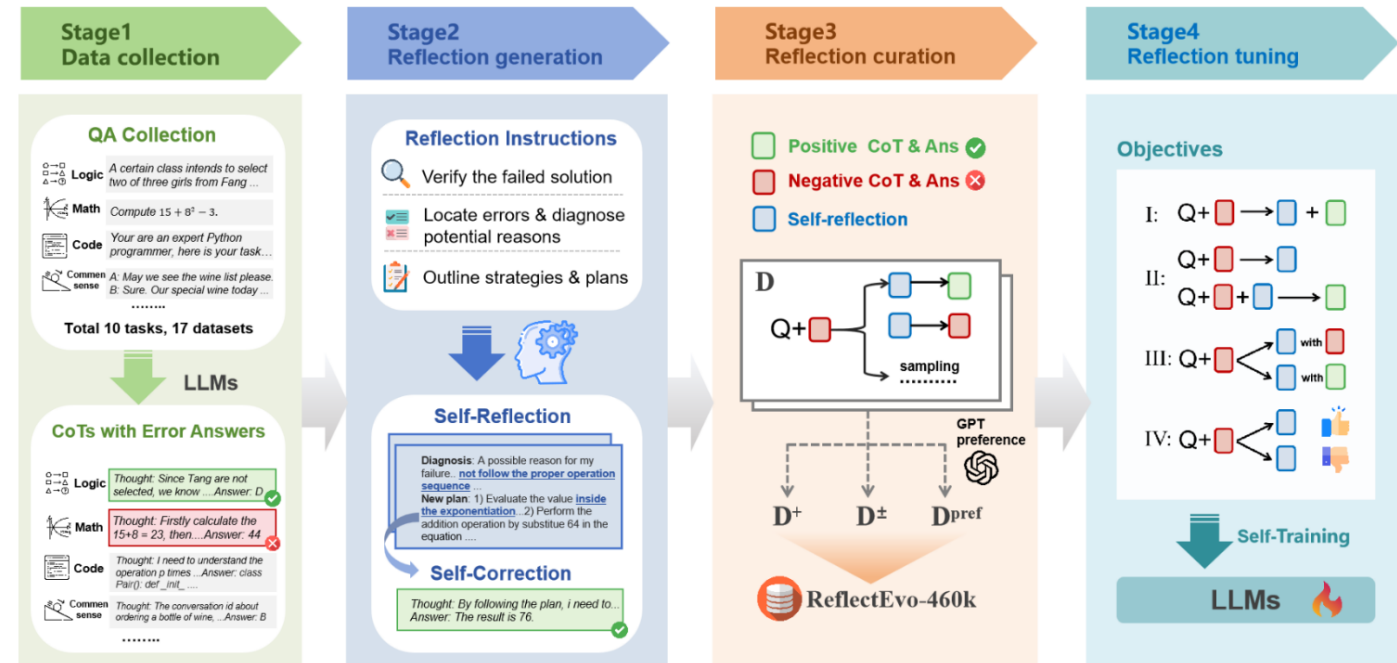


Stage 2 : Conflict-Aware Student Alignment. In this phase, we enable the student model to resolve potential conflicts which might arise because of multiple approaches to learning the same concept. We thus project all of them to a shared regularized latent space.

Self-Reflection in LLMs: frameworks & LatentEvolve

Why Self-Reflection?

- Recent works show that **LLMs can improve their own reasoning** by analyzing and refining their outputs.
- This creates a feedback loop where the model becomes both **solver and teacher**.
- Particularly effective for **small models**, where external data is limited.



Examples: R-Zero, Socratic-Zero, LatentEvolve, ReflectEvo.

Framework : LatentEvolve: Self-Evolving Test-Time Scaling in Latent Space

LatentEvolve: Self-evolving Test-Time Scaling

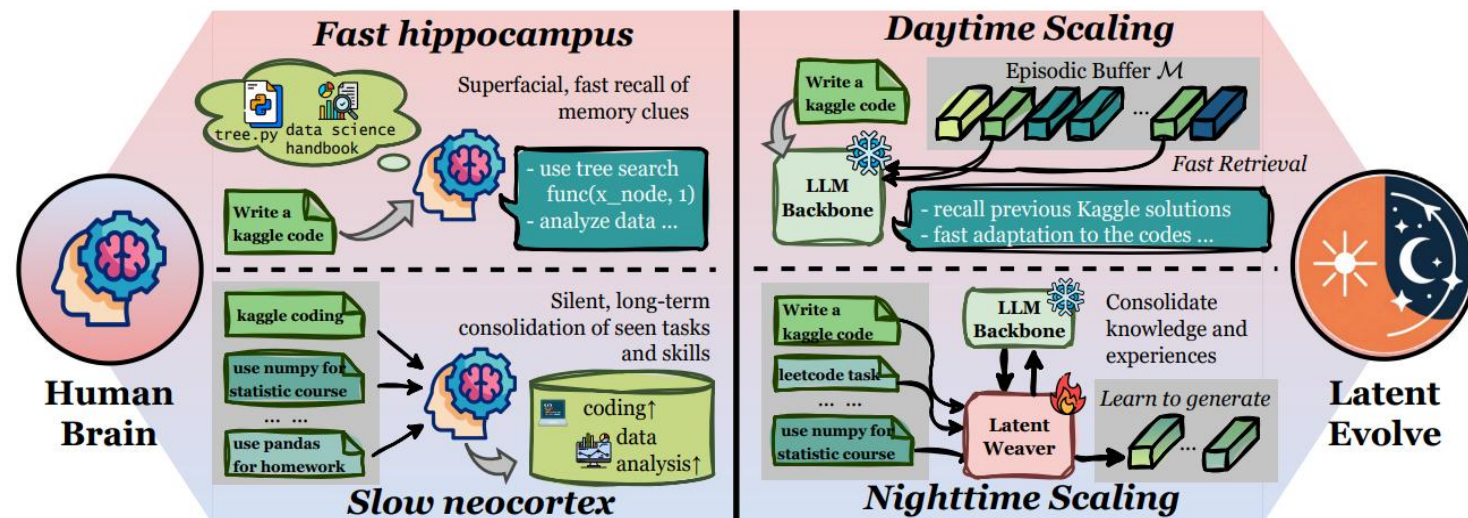
Core Idea: LLM improves its answer *during inference* by evolving latent representations over multiple iterations.

Two-phase cycle:

- **Daytime (Fast):** Retrieve best latent trajectories from memory → guide current decoding.
- **Nighttime (Slow):** Consolidate the improvements → update latent memory.

Why it's powerful:

- Works *without finetuning the large model*.
- Gains generalize across tasks/backbones.



Mimics Complementary Learning Systems.

Framework : LatentEvolve: Self-Evolving Test-Time Scaling in Latent Space

LatentEvolve Methodology Breakdown

Phase 1: Daytime Scaling (Fast System)

Objective: Optimize reasoning for the *current* specific query using past experiences.

- **Step 1: Associative Retrieval**
 - The system searches an **Episodic Buffer** for similar past problems.
 - Retrieves "triplets" of: (*Context*, *Base Latents*, *Optimized Latents*).
- **Step 2: Self-Supervised Refinement**
 - **Self-Rewarding:** The model evaluates its own reasoning path to guide the optimization.

```
Adding these up, the total cost of their order is:
$9 (Dakota's original order) + $4 (additional items by Ben) = $13.

Finally, to find the change they should receive from $15, we subtract the total cost from $15:
$15 - $13 = $2.

Therefore, the change they should receive from $15 is $2.<|im_end|>, -- Current Reward: -0.5
-- Loss: -7.0625
Verifier calculation_check approved the solution.
Verifier answer_correct disapproved the solution.
Verifier answer_completeness disapproved the solution.
Verifier understanding_check disapproved the solution.
Verifier Rule Format disapproved the solution.
-- New Answer: First solve the much change Dakota and Ben should receive from $15, let first to first the total cost
the15.1, we's break the initial of Dakota items items:
1 The cost of eggs is $3.
2 The cost of pancakes is $2.
3 The cost of 2 mugs of cocoa is $2 x 2 = $4.
4 The total cost of Dakota's meal is $3 + $2 + $4 = $9.
5 The tax is $1.
6 The total cost of Dakota's meal including tax is $9 + $1 = $10.
7 The total cost of Ben's meal is $10 + $2 = $12.
```

Framework : LatentEvolve: Self-Evolving Test-Time Scaling in Latent Space

LatentEvolve Methodology Breakdown

Phase 1: Daytime Scaling (Fast System)

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Step 1: Associative Retrieval

- The system searches an **Episodic Buffer** for similar past problems.
- Retrieves "triplets" of: (*Context, Base Latents, Optimized Latents*).

Step 2: Self-Supervised Refinement

- **Self-Rewarding:** The model evaluates its own reasoning path to guide the optimization, requiring no human labels.

Phase 2: Nighttime Scaling (Slow System)

- *Objective:* Train a small model with the extracted high-quality embeddings.

Process:

- Gathers all high-quality "Daytime" experiences (Output of Daytime-Scaling)
- Fine-tune the Latent Weaver on this data.
- The Weaver compresses these episodic memories into generalizable parameter weights.

Framework : LatentEvolve: Self-Evolving Test-Time Scaling in Latent Space

Learning by Teaching (LbT)

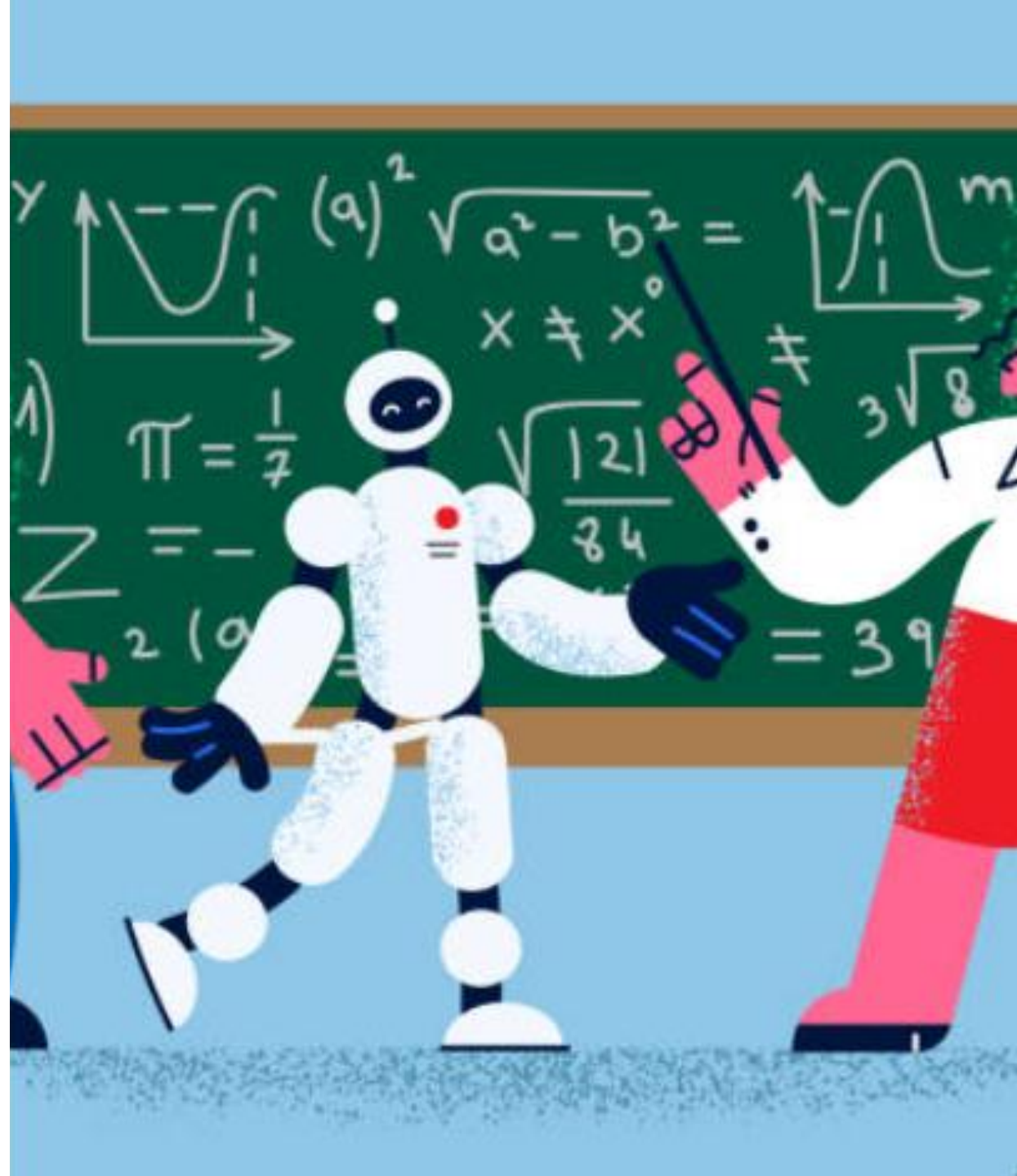
Beyond Distillation

Traditional methods rely on "Learning from Teachers" (e.g., Knowledge Distillation), transferring knowledge from strong to weak models.

The Feynman Technique

This paper asks: *Can LLMs improve their own reasoning by teaching others?*

Just as human teachers gain deeper understanding through the act of explaining concepts, this study explores if LLMs can enhance their own capabilities by generating teaching materials and observing student feedback.



Comparison of Paradigms



Learning from Teachers (LfT)

Direction: Strong $\rightarrow$ Weak

Mechanism: Knowledge Distillation / Imitation Learning.

Goal: Compress knowledge into smaller models.



Learning by Teaching (LbT)

Direction: Strong $\leftarrow$ Feedback $\leftarrow$ Weak

Mechanism: Teacher generates content, observes student performance, updates self.

Goal: Enhance the "Strong" model's reasoning.

Idea



- **Teach:** The model explains a concept.
- **Test:** A student model tries to use that explanation.
- **Critique:** If the student fails, we identify *why* the explanation was bad.
- **Train:** We update the model (DPO) to never make that explanation error again.
- **Self-Correction:** The pipeline creates a loop where the model autonomously improves its own training data.

Learning by Teaching (LbT)



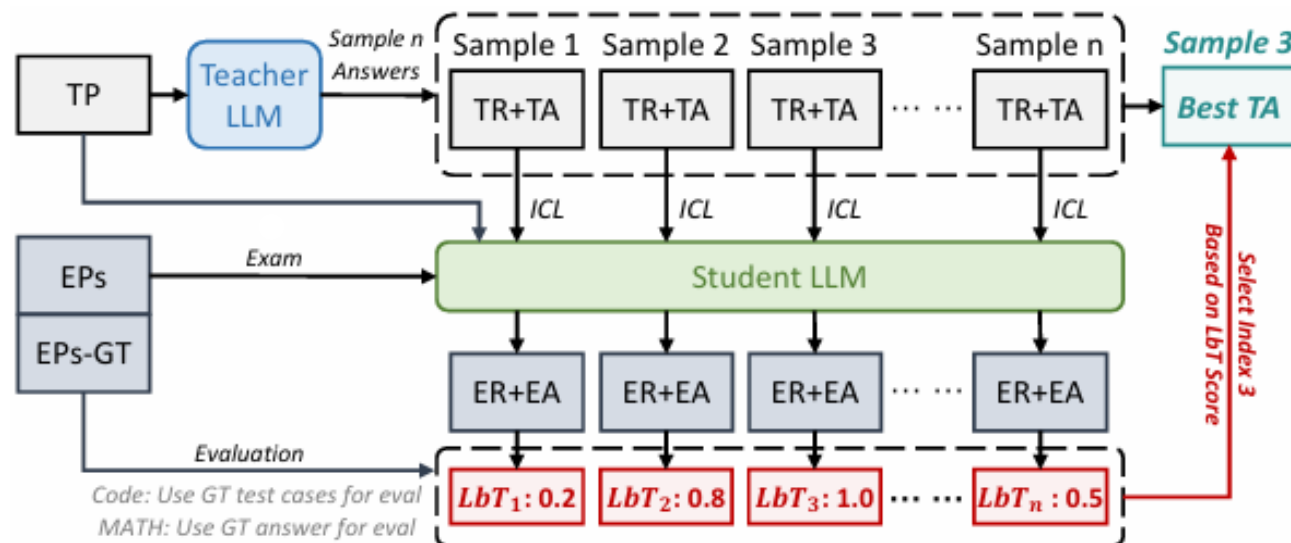
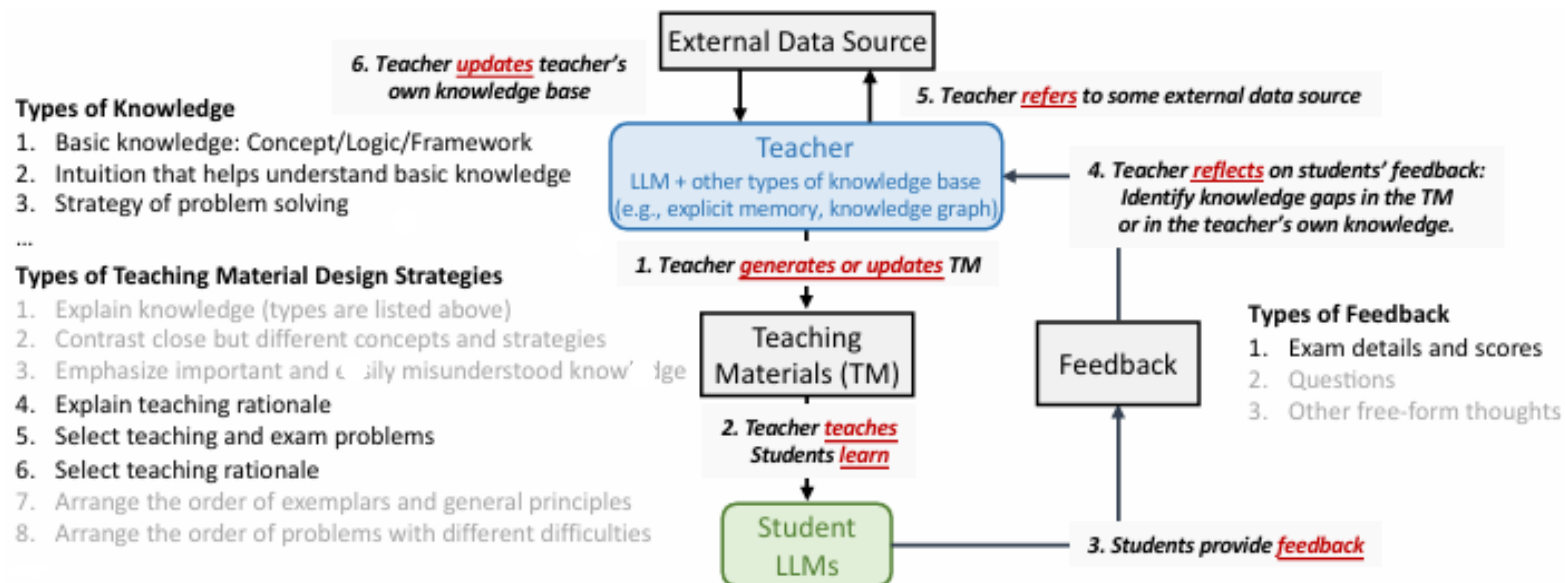
Teacher: Llama3.1
8B Instruct



Student: Llama3.1
8B Instruct



Judge: GPT OSS
20B



Ideal Tutor:

- Avoid Answer Leakage
- Use Socratic Questioning
- Provide Scaffolding
- Encourage Active Learning
- Promote long term Learning

This leads us to an Online, multi-turn RL framework

1. Traditional Supervised Methods

- **Prompt Engineering:**
Tedious to craft and maintain.
Sensitive to changes and prone to breaking.
- **Supervised Fine-Tuning (SFT):**
Requires scarce, high-quality dialogue data, making it expensive.
Generating synthetic data relies heavily on the model's current, potentially non-pedagogical capabilities.
Can suffer from severe generalization issues.

2. Reinforcement Learning from Feedback (RLHF)

- Typically relies on expensive, proprietary human preference annotations.
 - Requires training of a Reward model
-

Ideal Tutor:

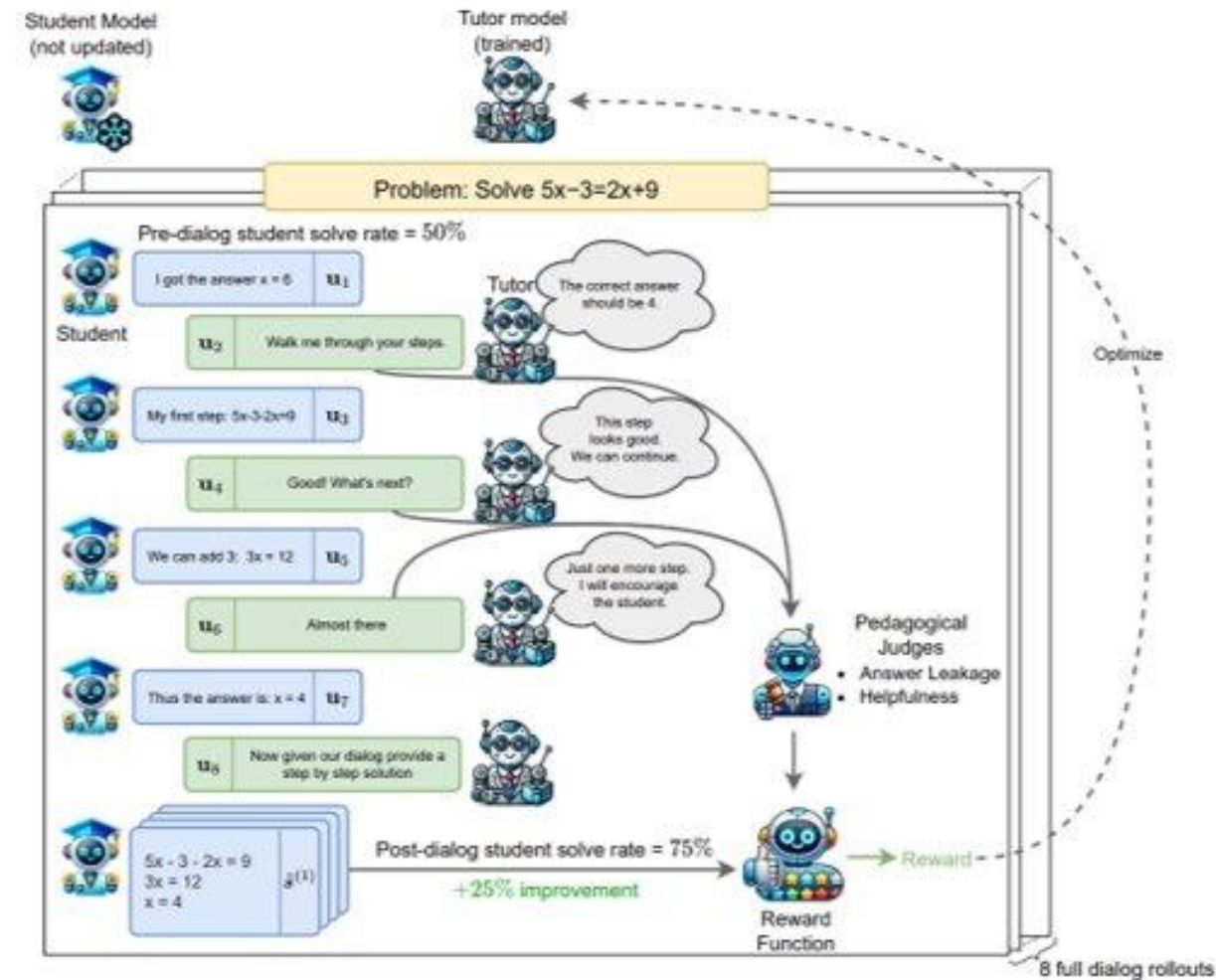
- Avoid Answer Leakage
- Use Socratic Questioning
- Provide Scaffolding
- Encourage Active Learning
- Promote long term Learning

- Want a Teacher model that must balance two objectives:

1) **Problem-solving competence** : Must be able to solve the problem correctly

2) **Pedagogical alignment** :
Must *communicate* that knowledge as helpful, stepwise guidance, without leaking the final answer.

- So we formulate this as a **multi-objective RL problem** (helpfulness vs answerleakage)



MODELS



Teacher: Llama3.1
8B Instruct



Student: Qwen2.5-
7B-Instruct



Judge: Qwen2.5-
14B-Instruct

Multi-turn conversation simulation phase

- For a single Q-A pair:
 - Student and teacher engage to create 8 different multi-turn conversations
- A conversation ends when either
 - Number of dialogues exceeds 15 dialogues
 - The teacher decides that it is done as student has solved the question

Reward Formulation

Total Reward (R) for a single conversation comprises of two terms

- $R = R_{\text{solution}} + \mu(R_{\text{pedagogy}} - 1)$

Finding R_{solution}

- After each conversation, we make the student solve the same question 8 times.
- Find out the solve rate and that is our R_{sol} for that conversation

Finding R_{pedagogy} :

- Each conversation is reviewed by a Judge LLM
- Judge checks if responses given by teacher follows predefined pedagogical rules
- Reward is 1 only if it satisfies ALL rules, else its 0
- If reward is zero, teacher is penalized with penalty μ
- Hyperparameter μ can be tuned to make teacher more strict or more helpful



AI

Hello! I am MentorX, ask me a question to begin.

Type your response here...



HALLUCINATION DETECTION AND MITIGATION AN OVERVIEW



A unified hallucination-mitigation pipeline for LLMs

The system implements a 4-stage FACTOR → HIDE → SEEK → PATCHER pipeline to detect, score, and repair hallucinations generated by Language Models without altering core model parameters.

The approach remains architecture-agnostic and training-agnostic, making it adaptable to instructional agents, reasoning tools, and general LLM deployments.



Representation-level detection before text-level verification

FACTOR analyzes internal latent geometry and identifies activation directions linked to stable vs. unstable reasoning modes.

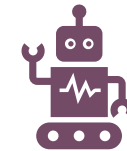
These extracted directions indicate where internal dynamics deviate from structure-aligned reasoning patterns.



Statistical dependence scoring via HSIC

HIDE evaluates the coupling between internal representations and truth-guided features using the Hilbert–Schmidt Independence Criterion.

Low HSIC ⇒ representational drift ⇒ high hallucination risk.



Multi-stage verification + controlled correction

SEEK uses multiple verification heads to assess factuality, solution validity, and reasoning stability.

When risk is high, PATCHER regenerates using controlled constraints and reference signals before re-verification or abstention.

HALLUCINATION DETECTION AND MITIGATION ARCHITECTURE



FACTOR Module

Extracts hidden states from all transformer layers.

Uses lightweight latent-factor analysis to isolate interpretable directions associated with unstable reasoning patterns.



HIDE (HSIC)

Aggregates multi-layer representations.

Computes drift indicators including HSIC, gating-entropy patterns, and mutual-information proxies.

Produces a continuous *representation-risk score*.



SEEK Verifier Ensemble

Multi-task heads predict:

- factual correctness,
- answer correctness,
- reasoning consistency.

An optional routing mechanism specializes on different error modes such as arithmetic slips or structural inconsistencies.



PATCHER

For uncertain or incorrect responses:

- invokes compact evidence retrieval,
- integrates verifier signals,
- regenerates with structured constraints,
- and re-checks before release.



Flow inside the generation pipeline

User query → LLM generates draft answer.

Hidden states → FACTOR + HIDE compute risk signals.

SEEK evaluates factual, answer-level, and reasoning integrity.

If flagged:

- evidence retrieval is triggered,
- PATCHER performs guided regeneration,
- SEEK checks again.

Final output or abstention.

HALLUCINATION DETECTION AND MITIGATION

KEY FEATURES, RESULTS, AND WHY IT MATTERS



Key Strengths of the Framework

Modular: Each module (FACTOR/HIDE/SEEK/PATCHER) is standalone and interchangeable.

Model-agnostic: Works with any LLM.

Transparent: Intermediate scores, evidence bundles, and verifier outputs make internal decisions observable.

Controlled correction: Unlike naive regeneration loops, PATCHER fires *only when needed*, preserving compute.



Results on Dataset from KGA+TMC

Factual Accuracy ↑ **+2.54 pts**

Answer Accuracy ↑ **+0.20 pts**

Reasoning Accuracy ↑ **+0.19 pts**

Calibration ECE ↓ **-2.55 pts** (improved confidence alignment)

HSIC signals show stronger coupling between hidden states and truth-aligned structure.



Why These Results Matter

Better calibration → reduces overconfident hallucinations.

Representation-level scoring + verifier-level scoring → multi-layer defense.

Evidence-supported regeneration stabilizes outputs and lowers error volatility.

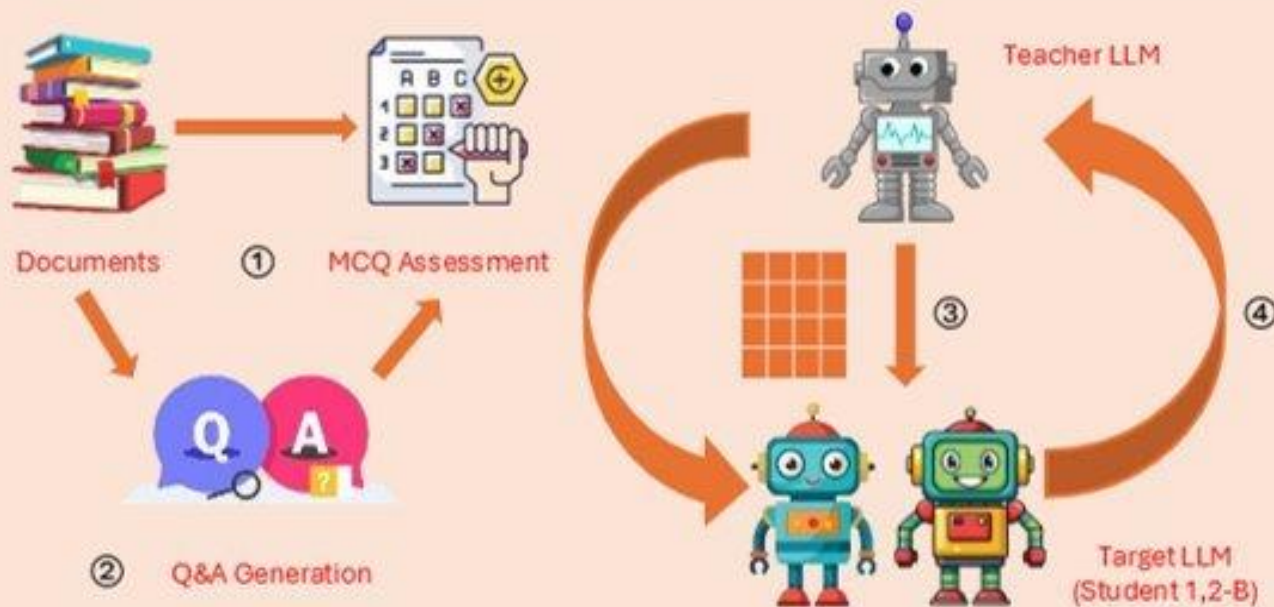
The architecture generalizes well because it targets representation dynamics rather than task-specific tricks.

THE END

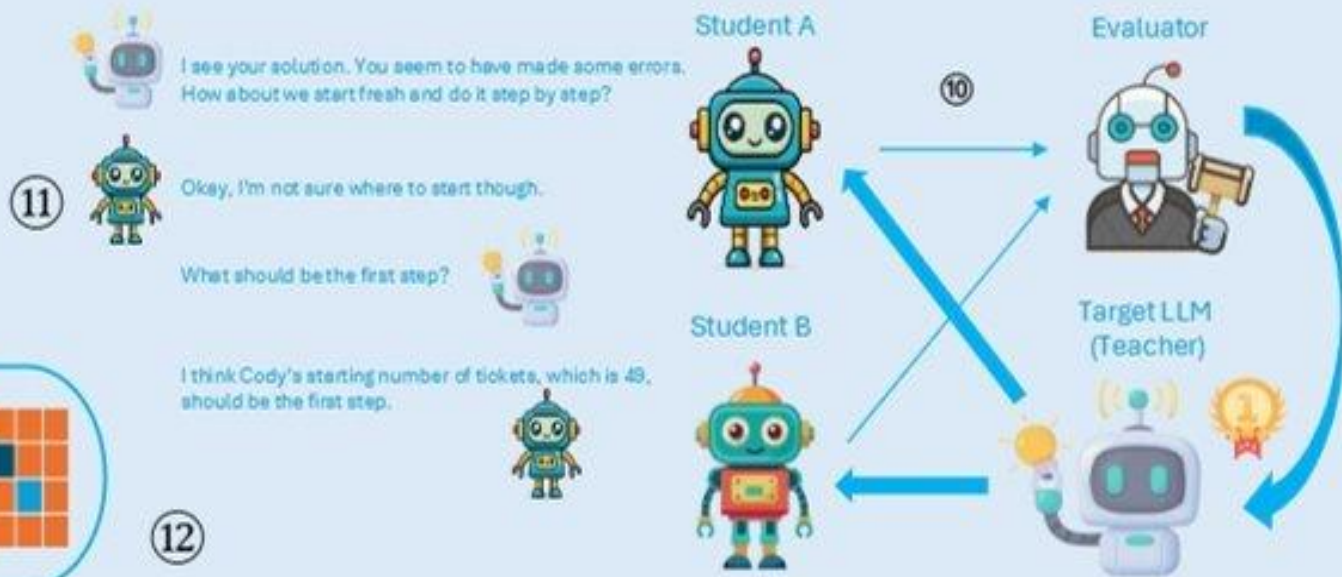
TEAM

Stage	Title	Member
1.	Teaching Material Curation	Kolipaka Bhargav Sashank
2.	Knowledge Gap Assessment	Kolipaka Bhargav Sashank
3.	Learning From Teacher	Y Ratna Sai Aakanksha
4.	Interactive Doubt Clearance	Y Ratna Sai Aakanksha
5.	Feedback Driven Practice	Ashwin K M
6.	Cumulative Learning Experience	Nitin Vetcha
7.	Post-Instructional Reflection	Shobhnik Kriplani
8.	Examination Feedback	Shobhnik Kriplani
9.	Learning By Teaching	Mehul Srivastava, Ashwin K M
10.	Socratic Styled Mentoring	Aditya Manjunatha, Sehaj Ganjoo
11.	Hallucination Mitigation	Kolipaka Bhargav Sashank
12.	User Memory	Nitin Vetcha, Ashwin K M

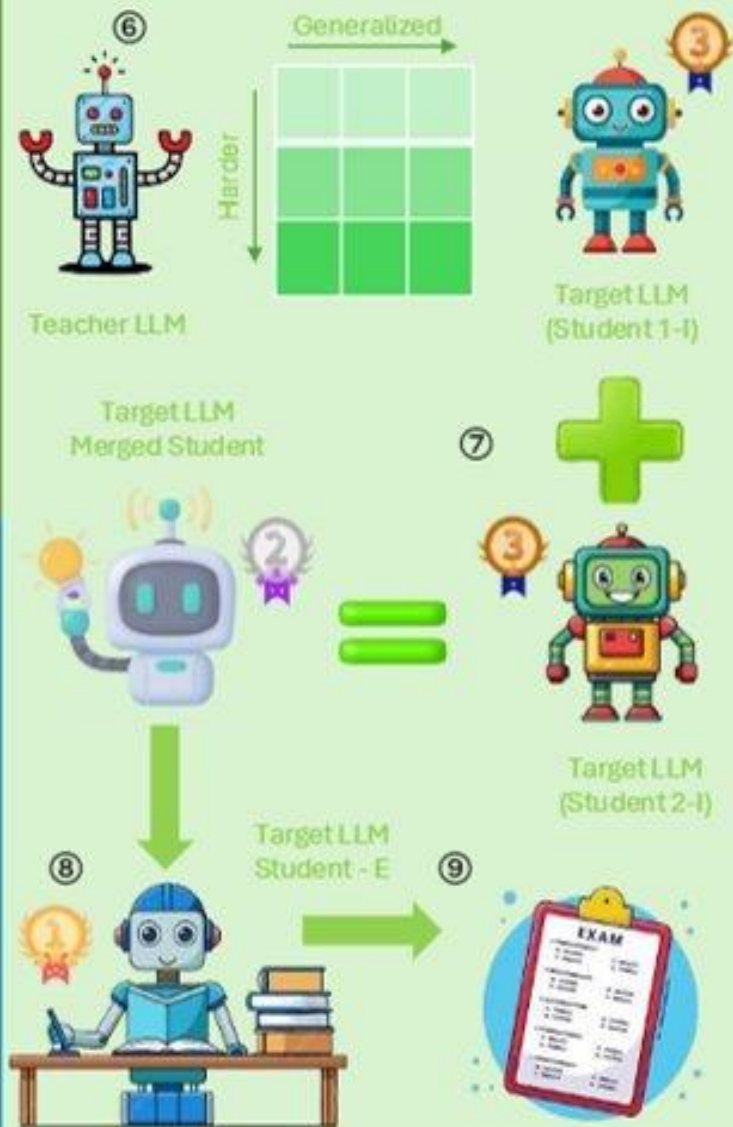
Stage I : Learning



Stage III : Teaching



Stage II : Mastery



Model Performance Comparison: Base vs. Adaptation Across Domains

